

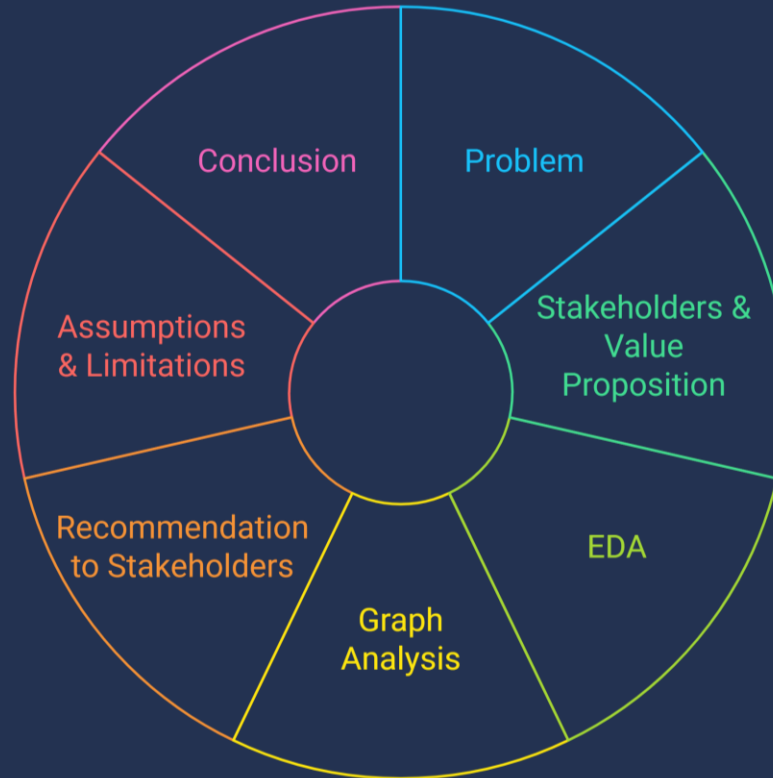


Institute_{of}
Data

Improving Student Outcomes: Analysing the Drivers Behind Rising Fail Rates

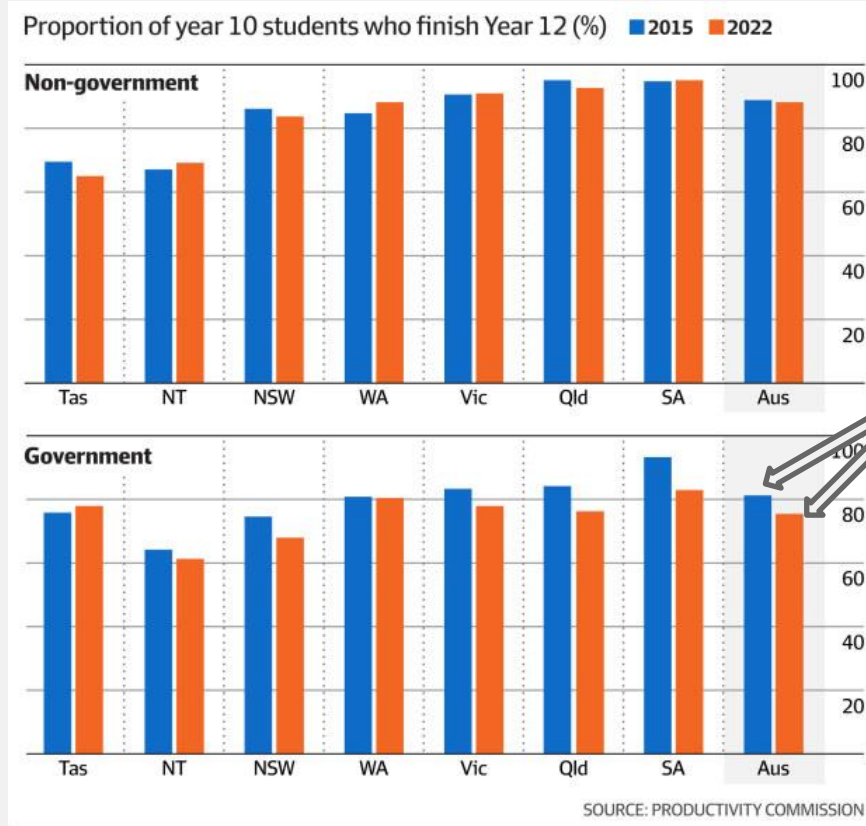
Jayden Doan
Mini Project 1 – Data Science

Project Overview (5min)

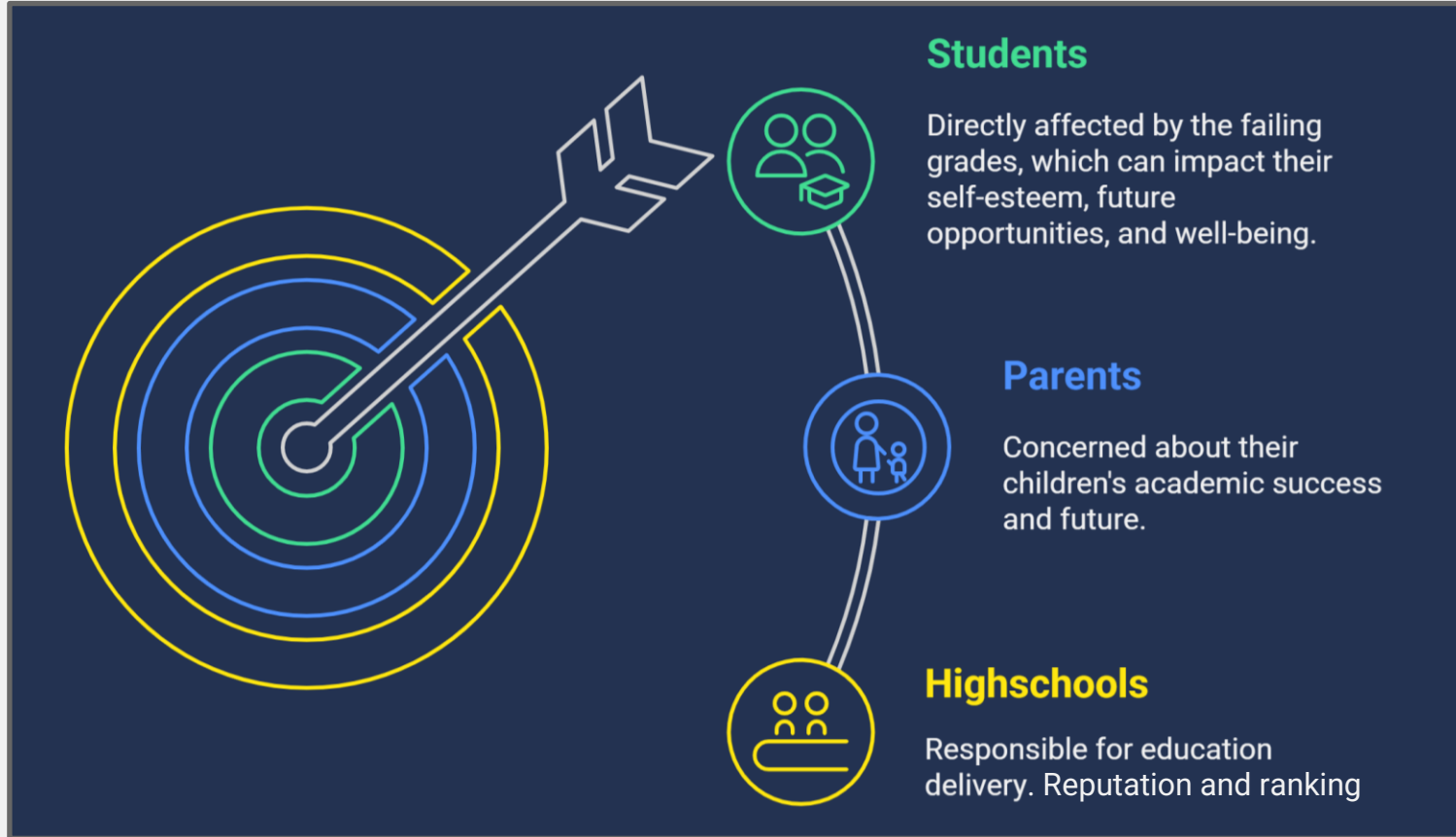


Problem

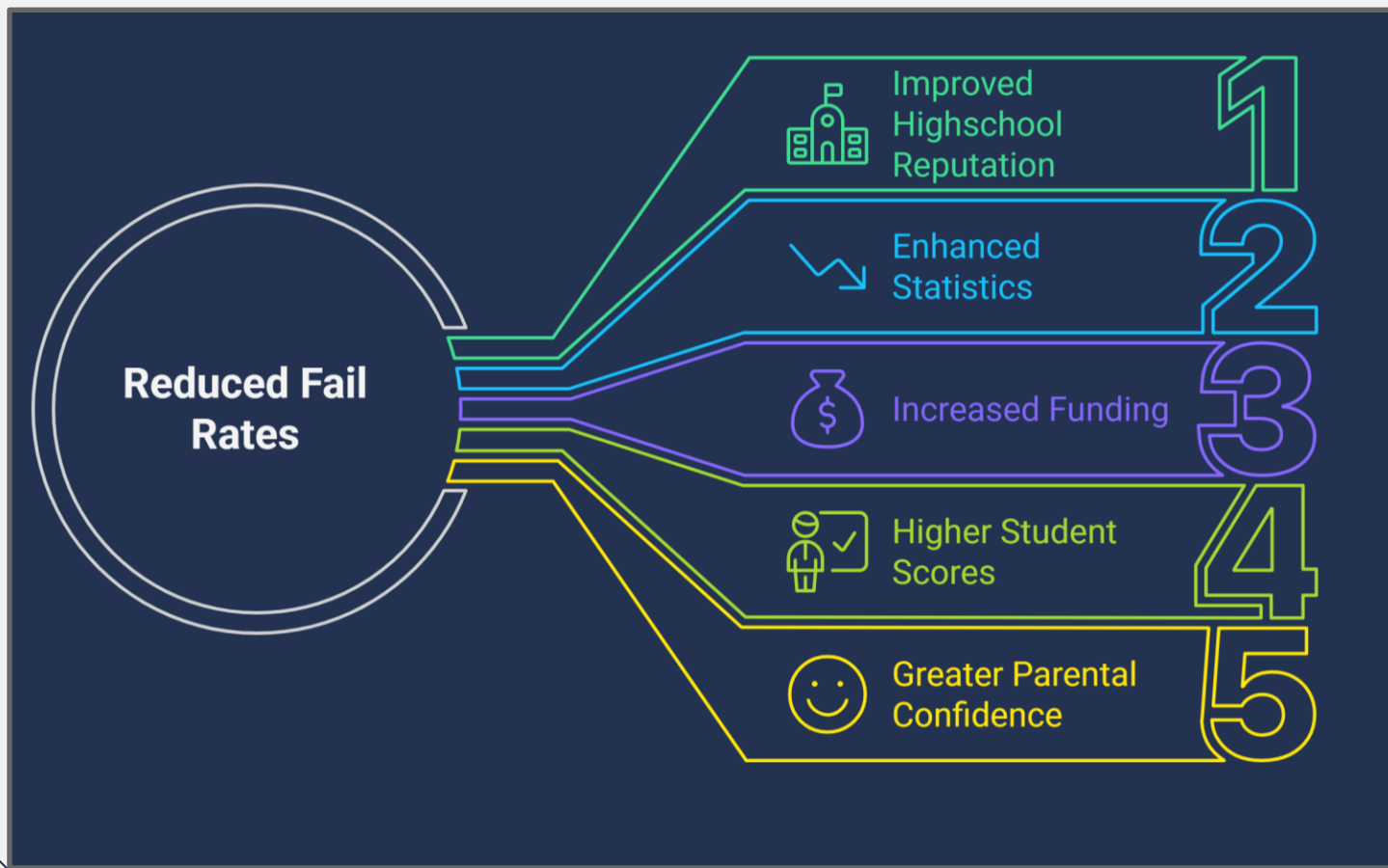
- High school students' fail rate has increased over the past few years (83.3% -> 79%)
- Understand how student performance is influenced and identify causes of lower scores to reduce the fail rate of students.



Who is affected? (Stakeholders)



Value Proposition



Exploratory Data Analysis (EDA)



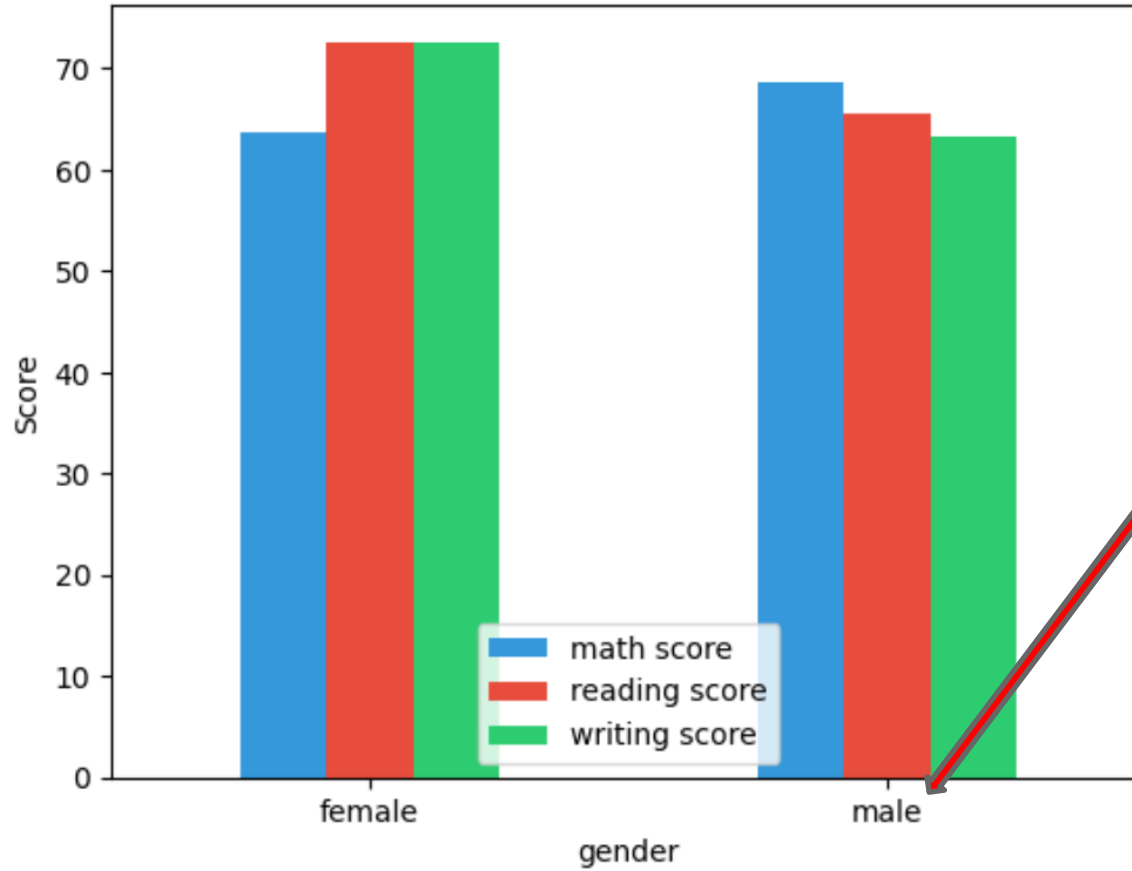
- Gender
- Race/ethnicity
- Parental level of education
- Lunch
- Test preparation course
- Math score
- Reading score
- Writing score

**Correlations with
average score?**

'StudentsPerformance.csv' – Students in high school from US (8 columns, 1000 rows)

<https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

Average Scores by Gender



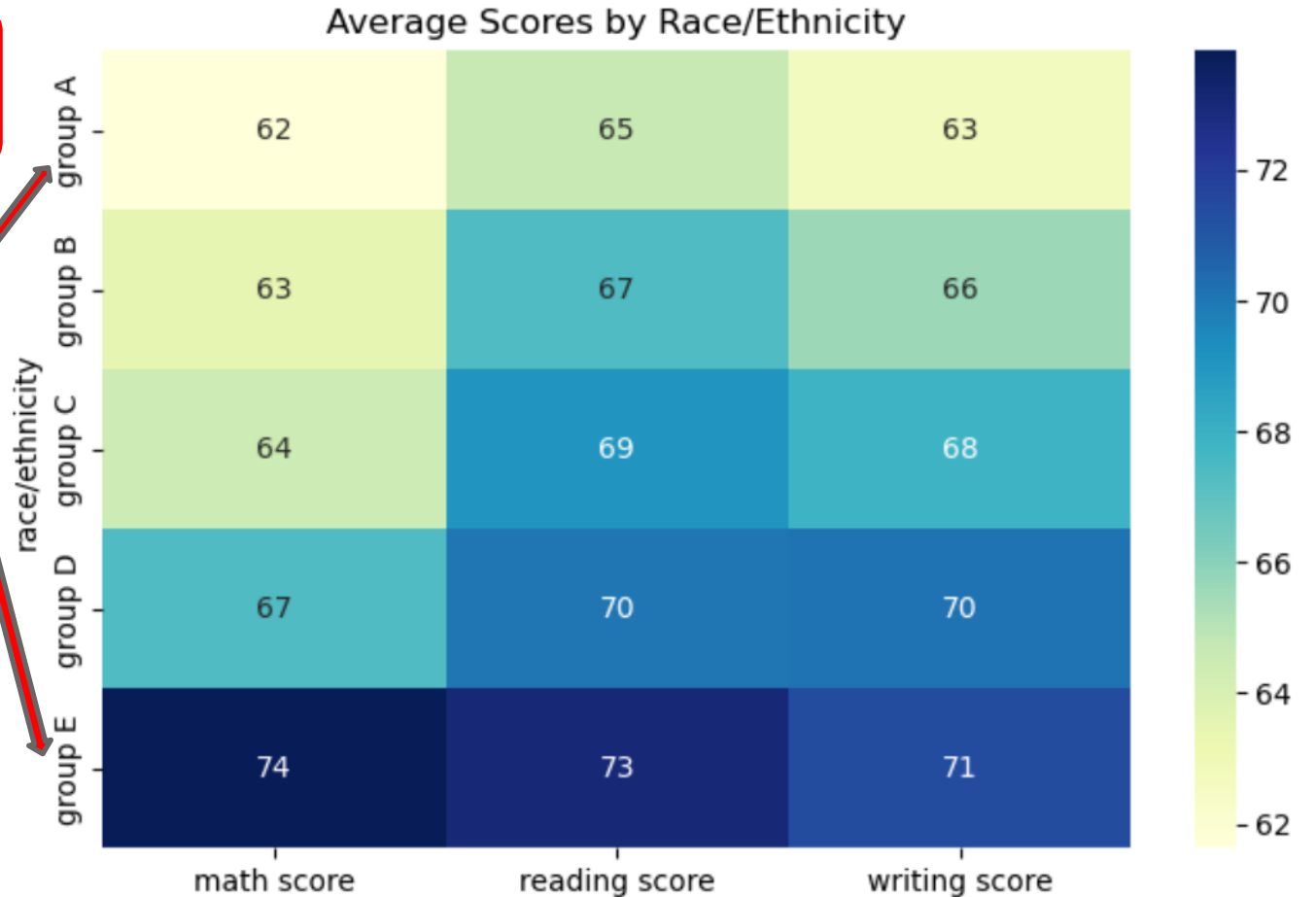
Males: math
Females: reading/writing

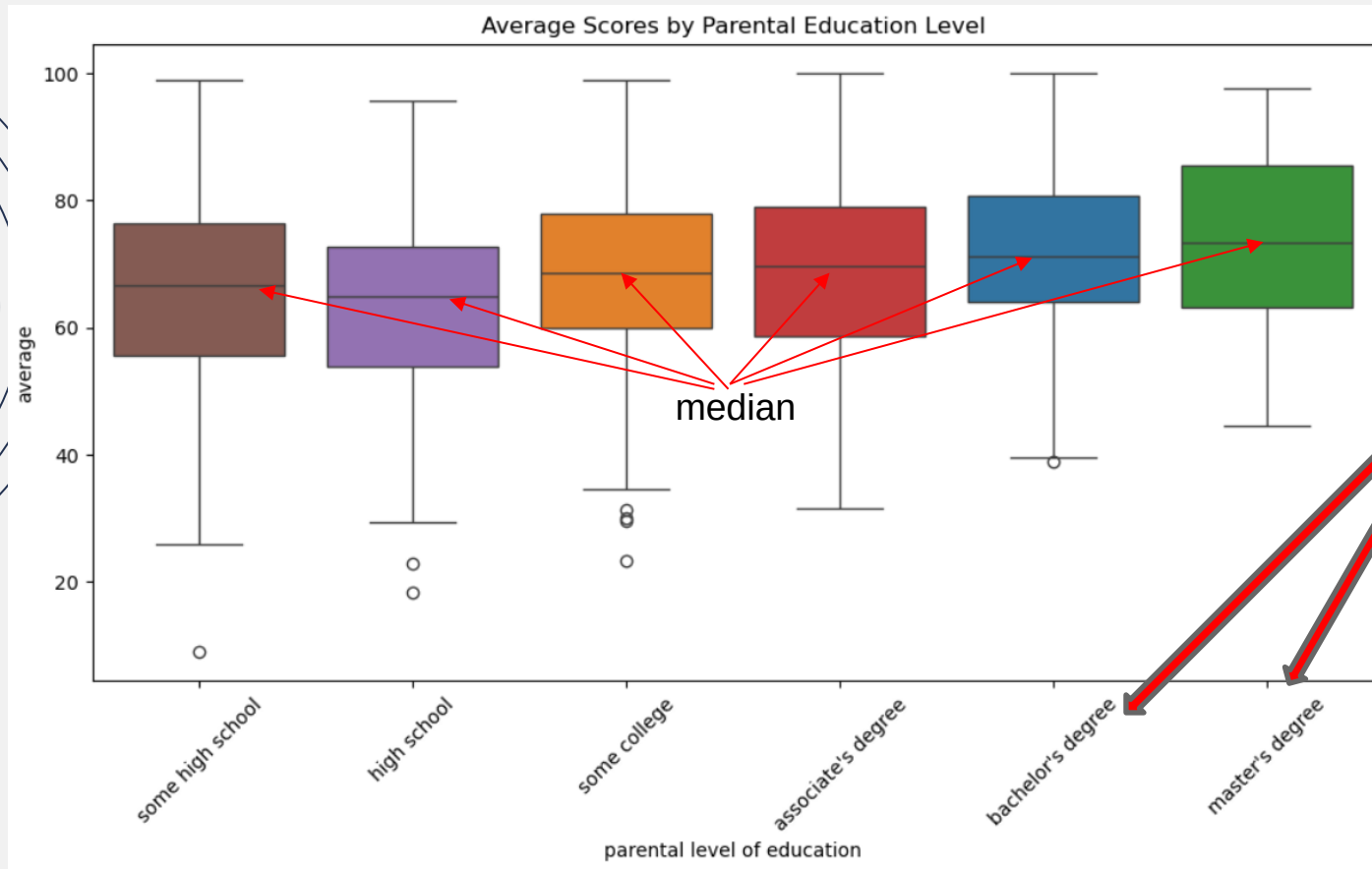
- Gender
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Heatmap

Group E scored highest. Clear correlation!

- Gender
- **Race/ethnicity**
- Parental level of education
- Lunch
- Test preparation course
- Math score
- Reading score
- Writing score

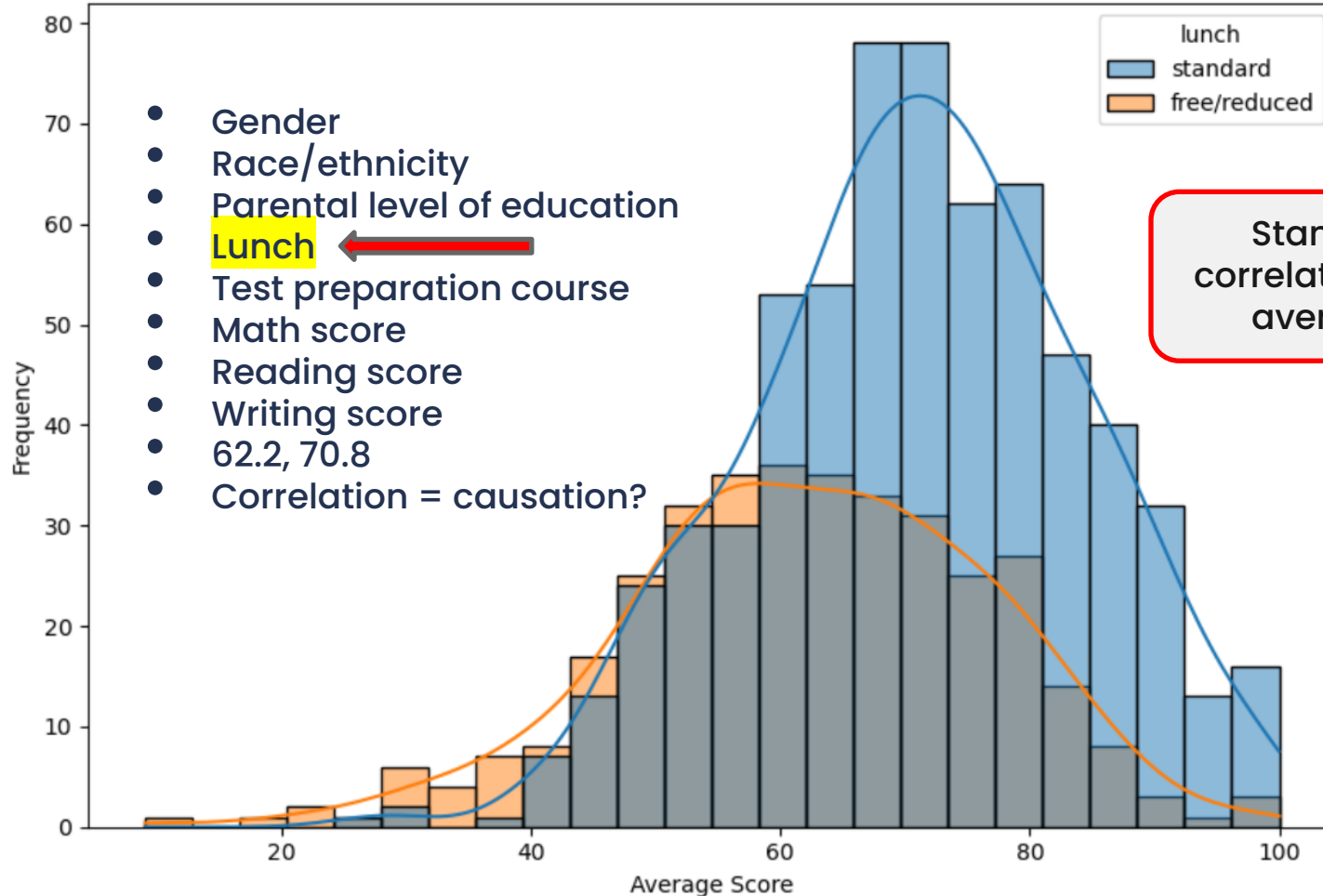




- Gender
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Higher parental education =
higher avg scores

Distribution of Average Scores by Lunch (Overall)

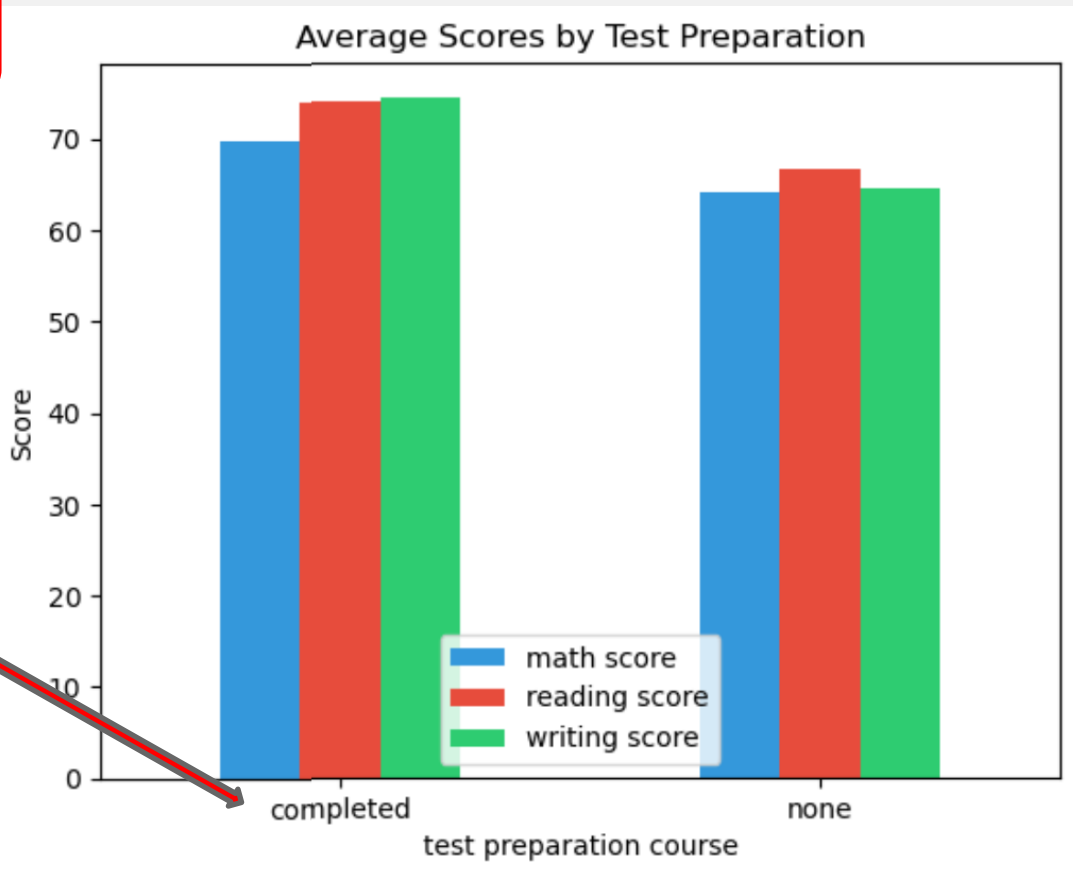


Test preparation course



Test prep
completed ->
higher score

- Gender
- Race/ethnicity
- Parental level of education
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- Writing score

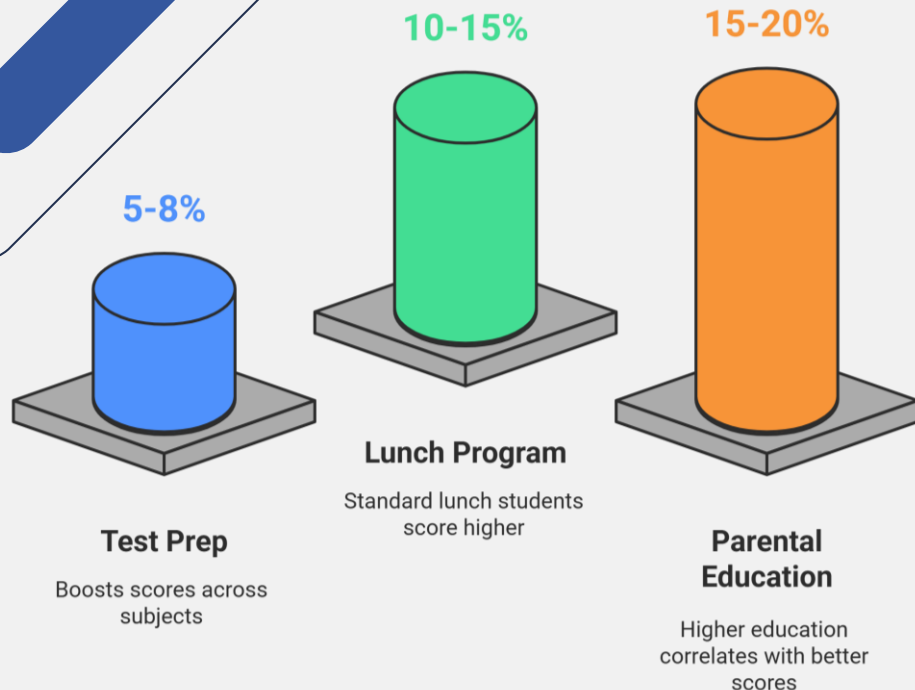


Positive correlations
with average score

Encoded to 0/1 or
1/2/3/4/5/6



Conclusions



Equity-focused interventions—(targeted test prep, lunch programs = more support to disadvantaged groups)—to bridge performance gaps, but better data is needed to address root causes.



Recommendations



Test Preparation

Encourage more students to take the test preparation course



Parental Engagement

Develop programs to educate parents on supporting their children's learning



Lunch Program

Investigate why students with free/reduced lunch performed worse. If it is related to nutrition, then make standard lunch more convenient/affordable. Trial standard lunch for free?



Assumptions & Limitations

Assumptions:

- test preparation course is equally accessible, equally effective to all students
- 'test preparation course' directly caused higher scores
- parental level of education is correlated with the parent's ability to support the child's education
- Socioeconomic status of parents is irrelevant

Limitations:

- dataset lacks information (school quality, teaching methods, student attendance, socioeconomic status...)
- Hard to conclude correlation = causation
- performance gaps = student factors or school quality?

Additional Data:

school resources, teacher qualifications, student attendance, school, income levels for deeper analysis





Code review (2min)

